

RIYA ROSE ROY



Electronics and Communication Engineering Student

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Career Objective

Electronics and Communication Engineering student passionate about transforming ideas into intelligent technology solutions through the fusion of AI/ML, embedded systems, and application development. Skilled through hands-on experience in building real-world projects, with a strong curiosity for designing smart, human-centered systems that blend hardware intelligence with software innovation. Enthusiastic about exploring emerging technologies, continuously learning through practical implementation, and developing impactful solutions that are both creative and future-oriented.

Education

NSS College of Engineering, Palakkad Expected May 2027

B.Tech in Electronics and Communication Engineering

CGPA: **6.98/10.00**

Carmel HSS, Chalakudy March 2023

Higher Secondary Education (Class XII)

Percentage: **97%**

St. Joseph's EMHS, Pazhookara March 2021

High School Education (Class X) – School Topper

Percentage: **97.6%**

Experience

Project Intern, IIST, Trivandrum May 2026 – June 2026

- Developed an autonomous robotic arm system using Deep Reinforcement Learning models.
- Optimized reward functions and data structures to minimize path-finding defects and maximize execution accuracy.

ICT Academy of Kerala May 2025 – June 2025

- Built an AI-based EV Range Predictor project.
- Worked on AI models using Linear Regression, Logistic Regression, Decision Tree, and Random Forest algorithms.

Positions of Responsibility

- **ENCORE Coordinator, IEEE SIGHT Kerala Section** (2026 – Present)
Conducted ENCORE sessions across regional campuses.
- **Secretary, IEEE SIGHT AG NSSCE** (2024 – Present)
Organized the NEXOTECH IEEE Conference 2026 and spearheaded Robocamp Jr.
- **IPR Lead, IEDC NSSCE** (2025 – Present)
Coordinated state-level YIP 7.0/8.0 campaigns and represented the campus at IEDC Summit 2025.

Projects

AI Fish Tank August 2025

- Engineered an interactive virtual environment with real-time gesture tracking and low-latency rendering using deep neural networks.
- Simulated fish behaviors including seek, flee, wander, and idle movements.
- **Tools Used:** Python, OpenCV, MediaPipe, Pygame, NumPy, gTTS

SHE WANTS – Women-Friendly Platform February 2026

- Built a full-stack health web application using a secure calculator-masked frontend layout to safeguard user

privacy.

- Developed a women-friendly platform integrating period tracking and emergency support systems.
- **Tools Used:** Java, Spring Boot, Python, Django, React, JavaScript, SQL, Google Maps API

8-Bit Microprocessor Design

April 2026

- Simulated an 8-bit microprocessor architecture featuring a modular Control Unit, Register File, and 24+ operation ALU.
- Verified processor functionality using Vivado simulations.
- **Tools Used:** Verilog HDL, Xilinx Vivado, Digital Logic Design, Behavioral Simulation

Smart Automated Delivery Box

April 2026

- Developed an IoT-enabled automated delivery box system.
- Implemented OTP verification, UART communication, and Telegram notifications.
- **Tools Used:** Python, Embedded C, UART, Telepot API, IoT Protocols, Hardware-Software Integration

MOODIFY

October 2025

- Developed an Arduino UNO-based mood detection system.
- Programmed Embedded C for mood classification using threshold analysis.
- **Tools Used:** Arduino IDE, Embedded C, Sensors, Serial Monitoring

Certifications

- Joy of Computing Using Python – NPTEL (Silver Badge), 2025
- Deep Learning: Foundations and Concepts – NPTEL, 2026

Technical Skills

Languages: Python, C, Embedded C

Technologies: AI/ML, OpenCV, MediaPipe, Flask, Arduino, ESP32

Tools: Xilinx Vivado, VS Code, Arduino IDE, Jupyter Notebook

Domains: Application Development, IoT, Computer Vision

Soft Skills

Leadership, Communication, Critical Thinking, Problem Solving, Collaboration, Time Management

Additional Information

Languages Known: English, Tamil, Malayalam

Interests: Movies, Reading, Travelling

Achievements

- Competed in national-level hackathons including Bharatiya Antariksh Hackathon 2025, Agentic AI Day (Google Cloud) 2025, and Cognizant Technoverse 2026.
- Presented the Smart Automated Delivery Box System at the YIP 8.0 District Evaluation Panel.